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CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

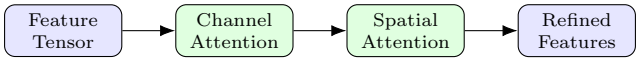
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

Which spatial features are related to other spatial features?

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50



Figure 1: Representative input images from the five flower classes: daisy, dandelion, roses, sunflowers, and tulips.

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

0.1 Attention Insertion Positions

Table 1: Attention positions and corresponding feature dimensions.

Position	Stage	Resolution	Channels
P1	After `conv2_x`	56×56	256
P2	After `conv3_x`	28×28	512
P3	After `conv4_x`	14×14	1024
P4	After `conv5_x`	7×7	2048

The principal position experiments focus on P2, P3 and P4. P1 retains higher-resolution spatial information but may require greater computational resources.

0.2 Feature Dimensions Before and After Attention

Let the input feature tensor be

$$F \in \mathbb{R}^{H \times W \times C}.$$

The refined feature representation is

$$F_{\text{att}} = \mathcal{A}(F),$$

where $\mathcal{A}$ denotes the selected attention mechanism.
For dimension-preserving attention modules,

$$\text{shape}(F_{\text{att}}) = \text{shape}(F).$$

Table 2: Expected feature dimensions before and after attention.

Position	Input Shape	Output Shape
P1	$(56, 56, 256)$	$(56, 56, 256)$
P2	$(28, 28, 512)$	$(28, 28, 512)$
P3	$(14, 14, 1024)$	$(14, 14, 1024)$
P4	$(7, 7, 2048)$	$(7, 7, 2048)$

The recorded P3 feature dimensions remained $(14, 14, 1024)$ before and after attention. For a single input image, the classifier produced an output of shape $(1, 5)$, corresponding to the five flower classes.

14. Main Experiment 1: Attention-Type Ablation

14.1 Experimental Configurations

Table 3 presents the attention configurations evaluated at P3. Six models were considered, including the baseline, channel attention, spatial attention, channel–spatial attention, self-attention, and combined attention.

Table 3: Attention configurations evaluated at P3

Model	Configuration	Channel	Spatial	Self-Attention
A0	Baseline	–	–	–
A1	Channel Attention	Yes	–	–
A2	Spatial Attention	–	Yes	–
A3	Channel–Spatial	Yes	Yes	–
A4	Self-Attention	–	–	Yes
A5	Combined Attention	Yes	Yes	Yes

14.2 Results and Comparison

Table 4 compares the classification performance and computational cost of the evaluated attention mechanisms using accuracy, precision, recall, macro F1-score, parameter count, and inference time.

Table 4: Performance and computational cost of attention mechanisms

Model	Accuracy	Precision	Recall	Macro F1	Parameters	Time (ms)
Baseline	0.9347	0.9350	0.9332	0.9339	23597957	7.6947
Channel	0.9401	0.9409	0.9376	0.9389	23730117	7.8742
Spatial	0.9292	0.9275	0.9293	0.9283	23598056	7.9831
Channel–Spatial	0.9201	0.9196	0.9206	0.9194	23730216	7.9179
Self-Attention	0.9347	0.9360	0.9331	0.9343	24909958	8.2831
Combined (A5)	0.9274	0.9258	0.9283	0.9267	25042217	8.3501

14.3 Interpretation

Channel attention achieved the best classification performance, with an accuracy of 94.01% and a macro F1-score of 0.9389. This represents a 0.54 percentage-point improvement over the baseline accuracy of 93.47%.

Spatial attention and channel–spatial attention achieved lower accuracies of 92.92% and 92.01%, respectively. Self-attention produced the same accuracy as the baseline, while the combined A5 model achieved 92.74%. Furthermore, A5 had the largest parameter count and the longest inference time.

Overall, channel attention provided the strongest classification performance with only a modest increase in model size and inference time. Combining multiple attention mechanisms did not improve performance in these experiments.

15. Main Experiment 2: Attention-Position Ablation

15.1 Attention Positions

Attention was inserted at three different stages of ResNet50 to investigate how the position of attention affects classification performance and computational cost. The evaluated positions were P2, P3, and P4, corresponding to the outputs of the conv3_x, conv4_x, and conv5_x stages, respectively.

Table 5: Attention insertion positions in ResNet50

Position	Stage	Resolution	Channels
P2	After conv3_x	28×28	512
P3	After conv4_x	14×14	1024
P4	After conv5_x	7×7	2048

15.2 Experimental Results

Table 6 presents the classification performance and computational cost of attention applied at different network positions. Accuracy, precision, recall, macro F1-score, total parameter count, and inference time were used for comparison.

Table 6: Effect of attention position on classification performance

Position	Accuracy	Precision	Recall	Macro F1	Total Parameters	Time (ms)
P2	0.9111	0.9090	0.9151	0.9106	23631368	7.845
P3	0.9201	0.9196	0.9206	0.9194	23730216	7.671
P4	0.9220	0.9196	0.9222	0.9201	24124520	7.778

15.3 Interpretation

P2 achieved an accuracy of 91.11%, whereas P3 improved the accuracy to 92.01%. P4 achieved the highest accuracy of 92.20% and the highest macro F1-score of 0.9201 among the three positions.

The results indicate a gradual improvement in classification performance as attention is moved from P2 to P4. Applying attention at P4 allows the model to refine deeper feature representations with smaller spatial dimensions and a greater number of channels. However, this improvement comes with an increase in model size, as P4 has the largest parameter count of 24,124,520.

Overall, P4 provided the best classification performance, while P3 achieved the shortest inference time of 7.671 ms. Therefore, P4 is preferable when classification performance is the primary objective, whereas P3 offers a favorable trade-off between accuracy, model complexity, and inference time.

16. Main Experiment 3: Multi-Stage Attention Ablation

16.1 Experimental Configurations

To investigate whether applying attention at multiple stages improves classification performance, four configurations were considered. These configurations include attention at P3 only, P4 only, both P3 and P4, and the extended combination of P2, P3, and P4. The evaluated configurations are summarized in Table 7.

Table 7: Attention configurations for the multi-stage experiment

Configuration	Attention Position
C1	P3 only
C2	P4 only
C3	P3 and P4
C4	P2, P3, and P4 (extended experiment)

16.2 Recorded Results

Table 8 presents the recorded classification results for attention applied at individual positions and at the combined P3+P4 configuration. The available metrics include accuracy, macro precision, macro recall, and macro F1-score. Results that were not recorded are explicitly marked as unavailable.

Table 8: Performance comparison of multi-stage attention configurations

Position	Accuracy	Precision	Recall	Macro F1	Total Parameters	Time (ms/image)
P3	0.9201	0.9196	0.9206	0.9194	23730216	8.0030
P4	0.9220	0.9196	0.9222	0.9201	24124520	7.8894
P3+P4	0.9383	0.9377	0.9371	0.9373	24256779	8.1021

16.3 Interpretation

The P3+P4 configuration achieved the highest accuracy of 93.83% and macro F1-score of 0.9373. Compared with P3 alone, its accuracy increased by 1.82 percentage points. It also outperformed P4 alone by 1.63 percentage points in accuracy.

The improvement came with a modest increase in model size. P3+P4 used 24,256,779 parameters and recorded an inference time of 8.1021 ms per image. P4 alone had the shortest inference time, at 7.8894 ms per image, but its accuracy was lower at 92.20%.

17. Main Experiment 4: Attention-Ordering Ablation

17.1 Experimental Configurations

To investigate the effect of attention ordering on classification performance, two configurations were considered at position P3. The first configuration applies channel attention followed by spatial attention, whereas the second reverses this order. The evaluated configurations are presented in Table 9.

Table 9: Attention-order configurations at P3

Configuration	Attention Sequence
O1	Channel Attention → Spatial Attention
O2	Spatial Attention → Channel Attention

17.2 Experimental Results

Table 10 presents the recorded classification performance and inference time for the evaluated attention sequences. Accuracy, macro precision, macro recall, macro F1-score, and inference time were considered to assess the effect of changing the order of attention operations.

Table 10: Performance comparison of attention order at P3

Attention Order	Accuracy	Precision	Recall	F1	Total Parameters	Time (ms)
Channel → Spatial	0.9201	0.9196	0.9206	0.9194	23,730,216	7.8779
Spatial → Channel	0.9274	0.9257	0.9262	0.9258	23,730,216	7.8360

17.3 Interpretation

The Spatial → Channel arrangement achieved the best performance, with an accuracy of 92.74%, macro precision of 92.57%, macro recall of 92.62%, and macro F1-score of 92.58%. Compared with the Channel → Spatial arrangement, it improved accuracy by 0.73 percentage points and macro F1-score by 0.64 percentage points. Macro precision and macro recall also improved by 0.61 and 0.56 percentage points, respectively. Both configurations had the same total parameter count of 23,730,216, indicating that the performance difference was not due to an increase in model size. The Spatial → Channel arrangement also recorded a slightly lower inference time of 7.8360 ms per image, compared with 7.8779 ms. These results suggest that applying spatial attention before channel attention was more effective for this experiment, providing better classification performance with marginally lower inference time.

18. Main Experiment 5: Spatial Attention Kernel-Size Ablation

18.1 Experimental Configurations

The effect of spatial attention kernel size was evaluated using 3×3, 5×5, and 7×7 kernels. These configurations were compared to investigate how the kernel size influences classification performance and computational efficiency. Accuracy, macro F1-score, and inference time were used as the evaluation metrics.

18.2 Experimental Results

Table 11 presents the performance comparison of the three spatial attention kernel sizes. The results provide a basis for identifying the kernel size that offers the best balance between classification performance and inference time.

18.3 Interpretation

The 5×5 and 7×7 kernels achieved the highest accuracy of 93.10% and macro F1-score of 93.04%. Both configurations outperformed the 3×3 kernel by 0.36 percentage points in accuracy and 0.41 percentage points in macro F1-score.

Table 11: Performance comparison of spatial attention kernel sizes

Kernel Size	Accuracy	Macro F1	Inference Time (ms/image)
3×3	0.9274	0.9263	7.6773
5×5	0.9310	0.9304	7.7349
7×7	0.9310	0.9304	7.7956

However, the 7×7 kernel required the longest inference time of 7.7956 ms per image, compared with 7.7349 ms for the 5×5 kernel. The 3×3 kernel was the fastest at 7.6773 ms per image, but achieved lower classification performance.

Overall, the 5×5 kernel provided the best balance between classification performance and computational efficiency. It achieved the same accuracy and macro F1-score as the 7×7 kernel while requiring less inference time. Therefore, the 5×5 kernel was the preferred configuration among the three kernel sizes evaluated.

19. Evaluation of Additional Attention Mechanisms

19.1 Experimental Configurations

Additional attention mechanisms were evaluated to investigate their effectiveness in improving the classification performance of the baseline ResNet50 model. The evaluated configurations included Channel Attention, Spatial Attention, Channel–Spatial Attention, Self-Attention, Squeeze-and-Excitation (SE), Efficient Channel Attention (ECA), and the Convolutional Block Attention Module (CBAM), along with the baseline without attention.

The models were compared using accuracy, macro precision, macro recall, macro F1-score, total parameter count, trainable parameter count, and inference time. The complete comparison is presented in Table 12.

Table 12: Performance comparison of attention mechanisms

Mechanism	Acc.	Prec.	Recall	F1	Total Params.	Trainable Params.	Time (ms)
No Attention	0.9347	0.9350	0.9332	0.9339	23597957	14963717	7.5523
Channel	0.9401	0.9409	0.9376	0.9389	23730117	15095877	7.8073
Spatial	0.9292	0.9275	0.9293	0.9283	23598056	14963816	7.6747
Channel–Spatial	0.9201	0.9196	0.9206	0.9194	23730216	15095976	7.8358
Self-Attention	0.9347	0.9360	0.9331	0.9343	24909958	16275718	7.9642
SE	0.9310	0.9300	0.9312	0.9302	23730117	15095877	7.8236
ECA	0.9201	0.9191	0.9226	0.9201	23597960	14963720	7.8548
CBAM	0.9238	0.9246	0.9220	0.9228	23730216	15095976	8.1261

19.2 Interpretation

Channel Attention achieved the highest accuracy of 94.01% and macro F1-score of 93.89%. Compared with the baseline, it improved accuracy by 0.54 percentage points and macro F1-score by 0.50 percentage points. Its macro precision and recall were 94.09% and 93.76%, respectively.

The baseline achieved the lowest inference time of 7.5523 ms per image. Self-Attention matched the baseline accuracy of 93.47%, but had the largest total parameter count of 24,909,958 and an inference time of 7.9642 ms per image.

Among the additional attention mechanisms, SE achieved an accuracy of 93.10%, while ECA and CBAM achieved 92.01% and 92.38%, respectively. Spatial Attention achieved 92.92%, whereas Channel–Spatial Attention achieved 92.01%. These results demonstrate that adding attention mechanisms does not necessarily improve classification performance.

Overall, Channel Attention provided the strongest classification performance, whereas the baseline offered the shortest inference time. Therefore, Channel Attention is preferable when classification performance is the primary objective, while the baseline remains a useful choice when minimizing inference time is more important. The findings emphasize the need to evaluate attention mechanisms in terms of both predictive performance and computational efficiency.

19.3 Questions for Attention-Mechanism Comparison

The following questions examine the classification performance, model complexity, computational efficiency, convergence behavior, and feature-localization capabilities of the evaluated attention mechanisms.

1. How does classification accuracy vary across attention mechanisms?

Channel Attention achieved the highest accuracy of 94.01%, followed by the baseline and Self-Attention, both at 93.47%. SE achieved 93.10%, Spatial Attention 92.92%, and CBAM 92.38%. Channel–Spatial Attention and ECA each achieved 92.01%. These results demonstrate that classification performance depends on the attention mechanism employed.

2. **How does macro F1-score vary across attention mechanisms?**
   Channel Attention obtained the highest macro F1-score of 93.89%, followed by Self-Attention at 93.43% and the baseline at 93.39%. SE achieved 93.02%, Spatial Attention 92.83%, CBAM 92.28%, ECA 92.01%, and Channel–Spatial Attention 91.94%. Channel Attention therefore achieved the best results in both accuracy and macro F1-score.
3. **How many parameters are introduced by each attention module?**
   The total parameter count ranged from 23,597,957 for the baseline to 24,909,958 for Self-Attention. Channel Attention used 23,730,117 parameters, Spatial Attention 23,598,056, Channel–Spatial Attention 23,730,216, SE 23,730,117, ECA 23,597,960, and CBAM 23,730,216. Self-Attention had the largest parameter count among the evaluated models, indicating the greatest overall model size.
4. **How does each attention mechanism influence inference time?**
   The baseline achieved the shortest inference time of 7.5523 ms per image. Spatial Attention required 7.6747 ms, Channel Attention 7.8073 ms, SE 7.8236 ms, and Channel–Spatial Attention 7.8358 ms. ECA required 7.8548 ms, Self-Attention 7.9642 ms, and CBAM 8.1261 ms. Thus, all evaluated attention configurations had longer recorded inference times than the baseline.
5. **Does attention improve the convergence behavior?**
   The available classification metrics do not establish whether attention improves training convergence. This question requires a comparison of training and validation loss or accuracy curves across epochs under consistent training conditions. Such an analysis would help determine whether an attention mechanism accelerates convergence or improves training stability.
6. **Does attention produce more focused feature localization?**
   Classification metrics alone cannot confirm whether attention produces more focused spatial feature localization. This can be investigated using attention maps, spatial-attention overlays, or Grad-CAM visualizations generated for the same input images. These visualizations can help assess which image regions contribute to the model's predictions.
7. **Does an increase in attention complexity always improve classification performance?**
   No. Self-Attention had the largest parameter count but achieved 93.47% accuracy, which was lower than the 94.01% achieved by Channel Attention. Similarly, Channel–Spatial Attention achieved 92.01% accuracy. Therefore, increased architectural complexity did not consistently lead to better classification performance in the evaluated experiments.
8. **Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?**
   Channel Attention achieved the highest accuracy and macro F1-score with 23,730,117 parameters and an inference time of 7.8073 ms per image. Self-Attention used 24,909,958 parameters and required 7.9642 ms per image, while achieving lower accuracy. These results indicate that Channel Attention provided better classification performance with fewer parameters and shorter inference time than Self-Attention.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.
The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Table 13: Overall performance comparison of attention mechanisms

No.	Model	Position	Accuracy	Precision	Recall	Total Params.	Time (ms)
0	Baseline	–	0.9347	0.9350	0.9332	23597957	7.5673
1	Channel Attention	P3	0.9401	0.9409	0.9376	23730117	7.6675
2	Spatial Attention	P3	0.9292	0.9275	0.9293	23598056	7.6061
3	Channel–Spatial	P2	0.9111	0.9090	0.9151	23631368	7.8450
4	Channel–Spatial	P3	0.9201	0.9196	0.9206	23730216	7.6711
5	Channel–Spatial	P4	0.9220	0.9196	0.9222	24124520	7.7778
6	Self-Attention	P3	0.9347	0.9360	0.9331	24909958	7.8668
7	SE Attention	P3	0.9310	0.9300	0.9312	23730117	7.5926
8	ECA Attention	P3	0.9201	0.9191	0.9226	23597960	7.8403

The macro F1-scores are presented separately in Table 14 to make the values easier to read and compare across configurations.

Table 14: Macro F1-score comparison

Model	Macro F1-score
Baseline	0.9339
Channel Attention (P3)	0.9389
Spatial Attention (P3)	0.9283
Channel–Spatial (P2)	0.9106
Channel–Spatial (P3)	0.9194
Channel–Spatial (P4)	0.9201
Self-Attention (P3)	0.9343
SE Attention (P3)	0.9302
ECA Attention (P3)	0.9201

Interpretation

Channel Attention at P3 achieved the best overall results, with 94.01% accuracy and a macro F1-score of 93.89%, improving baseline accuracy by 0.54 percentage points. In contrast, Channel–Spatial Attention at P2 had the lowest performance, while P4 was the strongest of the three Channel–Spatial configurations. The baseline remained the fastest at 7.5673 ms per image, with SE Attention close behind. Overall, Channel Attention offered the best balance of classification performance and computational cost, showing that a more complex attention mechanism does not always lead to better results.

24. Graphical Performance Analysis

24.1 Training and Validation Accuracy

The training and validation accuracy curves show how model performance changes over epochs. A growing gap between the curves may indicate overfitting, particularly when training accuracy improves while validation accuracy stops improving or decreases.

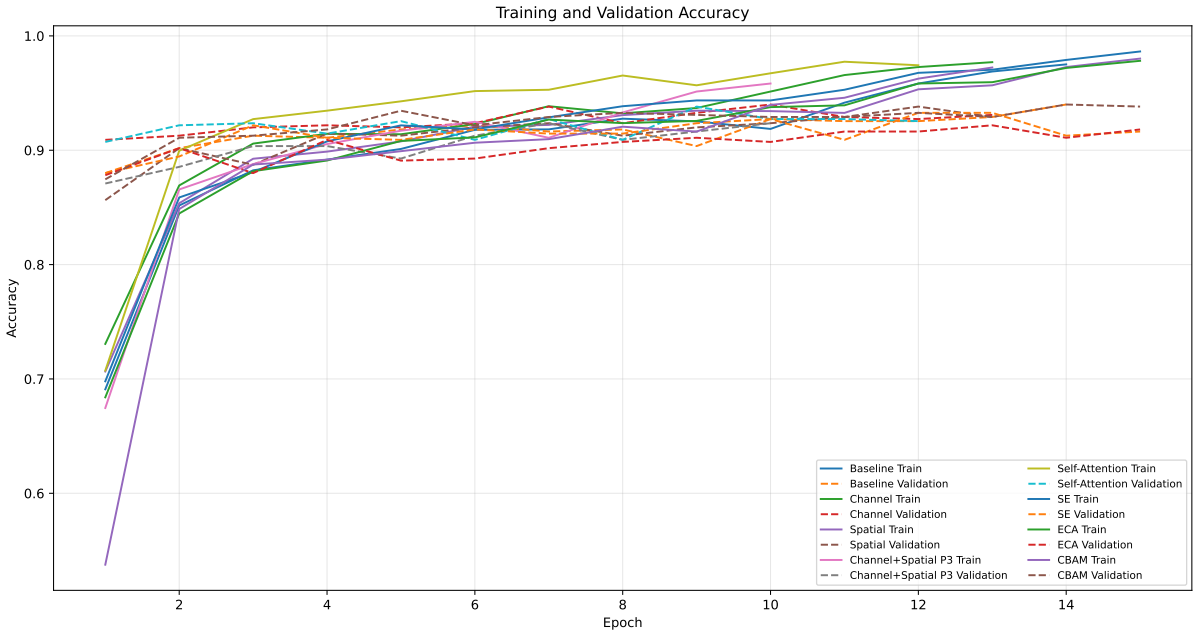


Figure 2: Training and validation accuracy across epochs.

24.2 Training and Validation Loss

Training and validation loss provide further insight into model learning and convergence. A decrease in both curves generally indicates that the model is learning effectively, while an increasing validation loss may suggest overfitting.

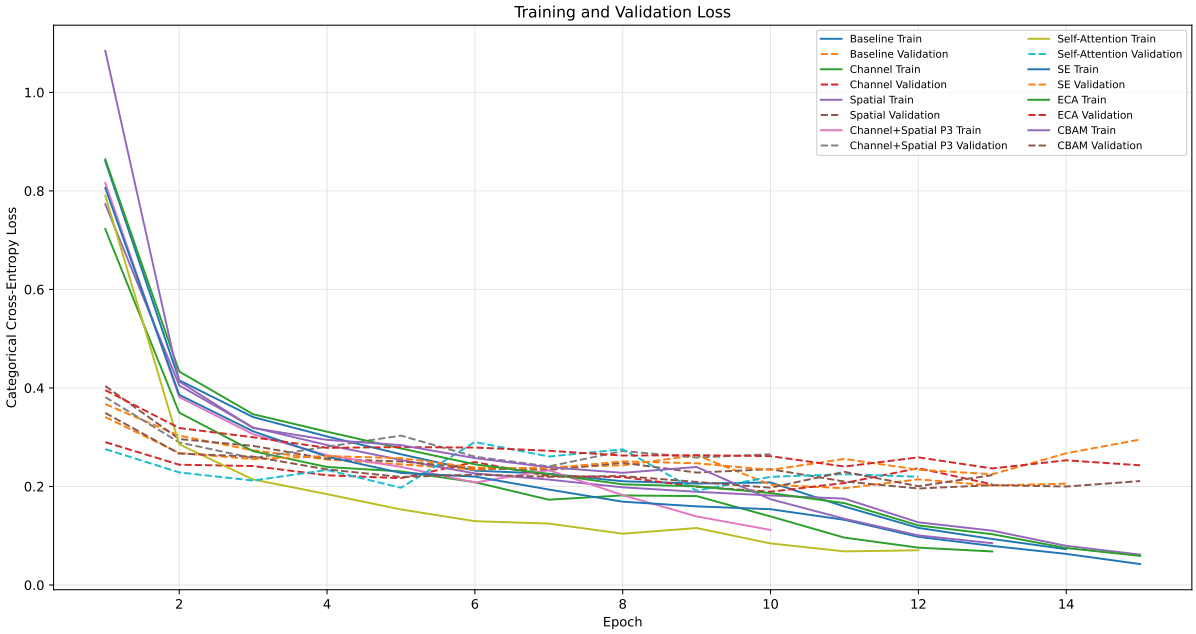


Figure 3: Training and validation loss across epochs.

24.3 Accuracy Comparison Across Attention Mechanisms

Channel Attention achieved the highest recorded accuracy at 94.01%, outperforming the baseline at 93.47%. The other evaluated attention mechanisms did not exceed the baseline in the recorded results, showing that adding attention does not always improve classification performance.

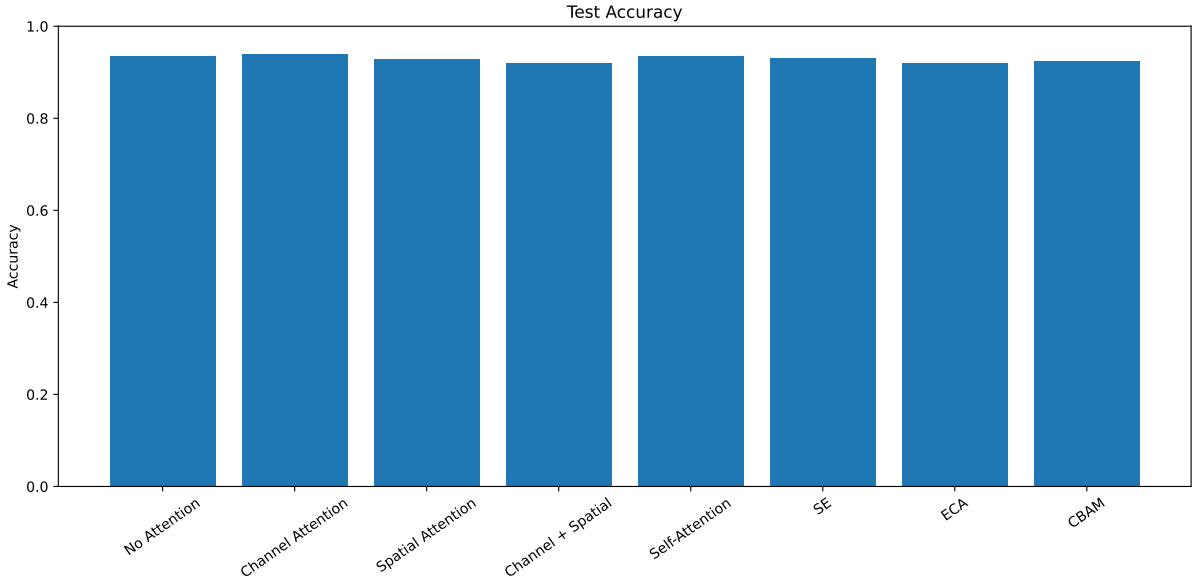


Figure 4: Test accuracy for different attention mechanisms.

24.4 Macro F1-Score Comparison

Channel Attention also achieved the highest macro F1-score of 0.9389, compared with 0.9339 for the baseline. This result supports its strong overall classification performance across the five flower classes.

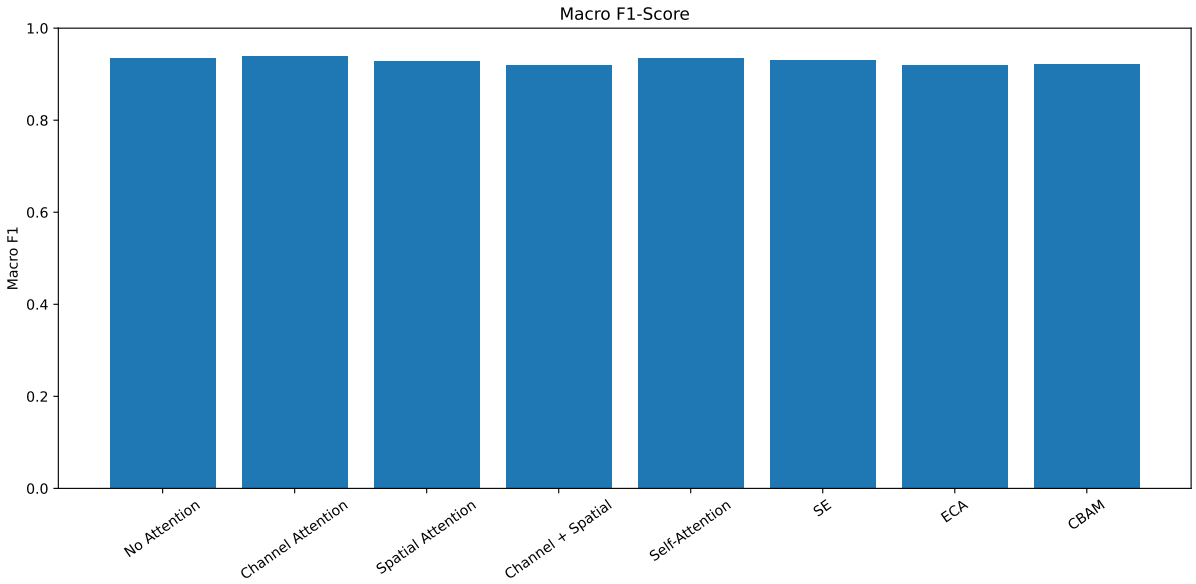


Figure 5: Macro F1-score comparison across attention mechanisms.

24.5 Accuracy and Model Complexity

The accuracy-versus-parameter plot compares classification performance with model size. Self-Attention had the largest parameter count among the compared mechanisms but did not outperform Channel Attention. This suggests that a larger model does not necessarily provide better classification results.

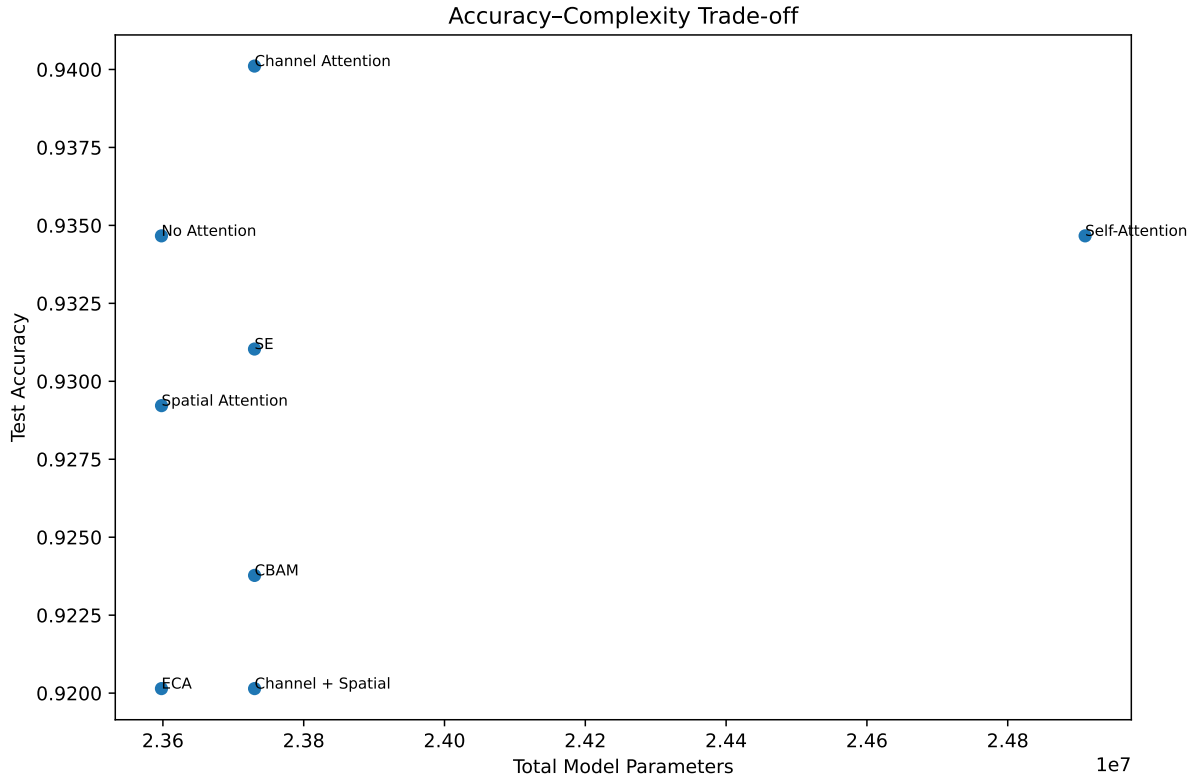


Figure 6: Relationship between test accuracy and total parameter count.

24.6 Macro F1-Score and Inference Time

The macro F1-score-versus-inference-time plot highlights the trade-off between classification performance and computational cost. Channel Attention achieved the highest macro F1-score, while the baseline had the lowest measured inference time. The preferred model therefore depends on whether classification performance or inference efficiency is the main priority.

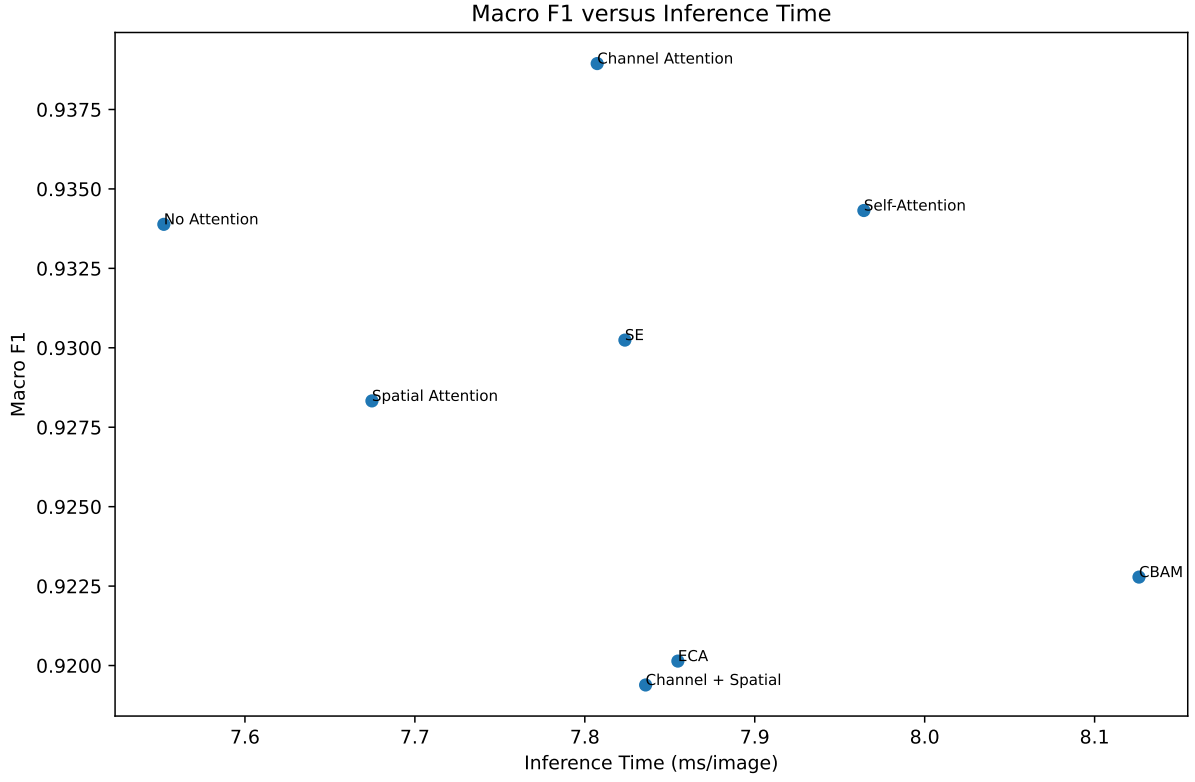


Figure 7: Relationship between macro F1-score and inference time per image.

25. Confusion Matrix Analysis

25.1 Baseline ResNet50

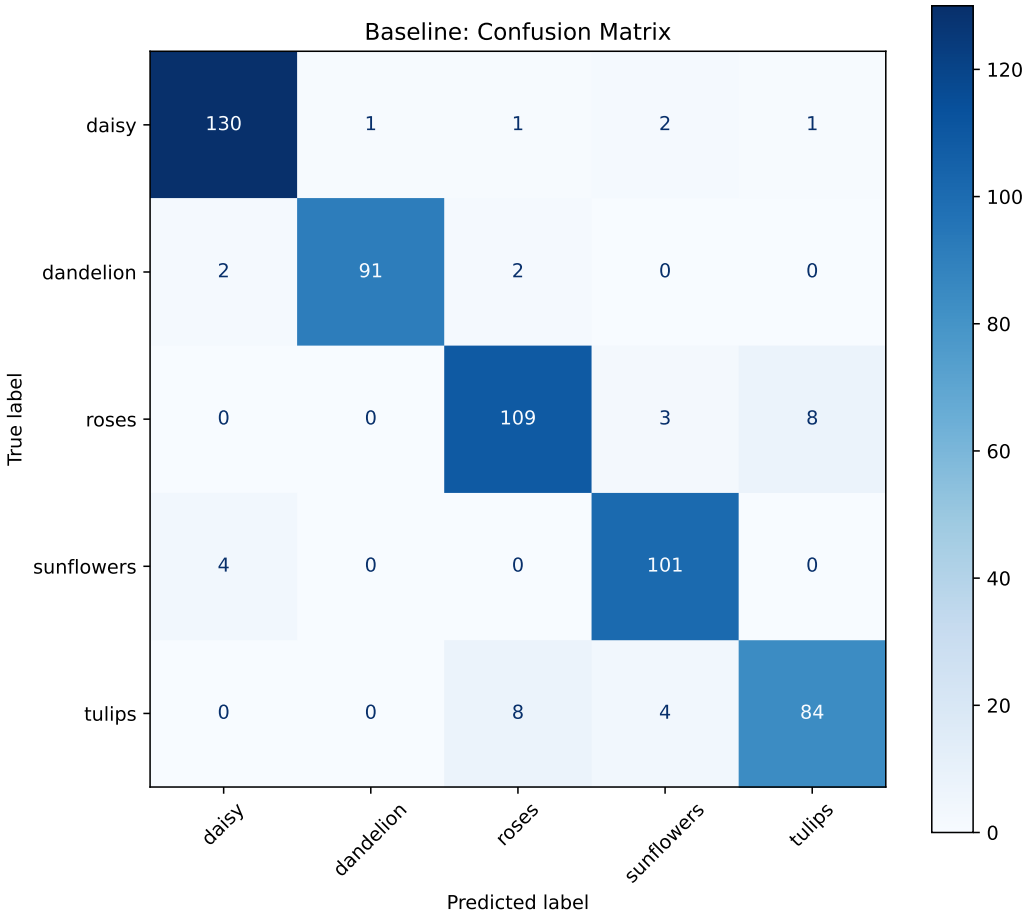


Figure 8: Confusion matrix for baseline ResNet50.

The baseline ResNet50 achieved 93.47% accuracy. It performed particularly well on dandelions and daisies, with F1-scores of 0.9733 and 0.9594, respectively. Tulips had the lowest F1-score (0.8889), making them the most challenging class for the baseline model.

25.2 Channel–Spatial Attention at P3

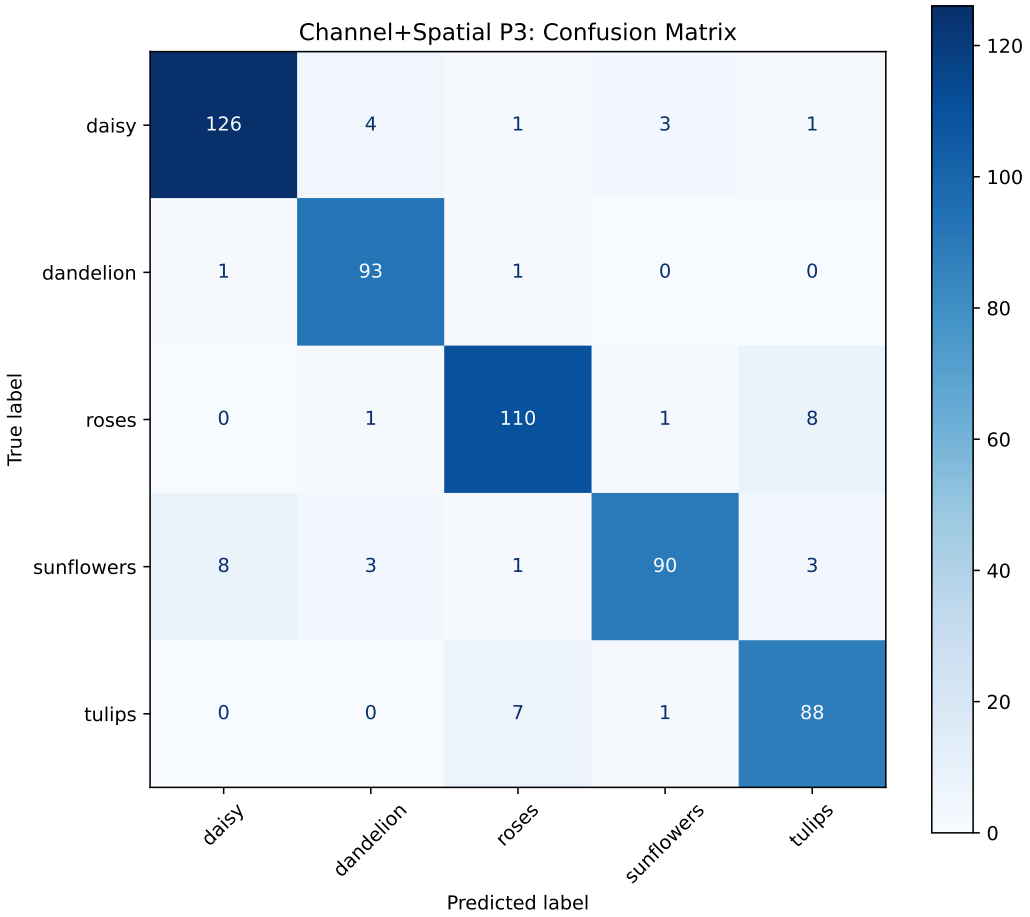


Figure 9: Confusion matrix for Channel–Spatial attention at P3.

Channel–Spatial attention at P3 achieved 92.01% accuracy, below the baseline. Dandelion recall increased from 0.9579 to 0.9789, while sunflower recall dropped from 0.9619 to 0.8571. This suggests that the attention mechanism helped some classes but made predictions less reliable for others.

25.3 Class-Wise Metric Comparison

Table 15 compares precision, recall, and F1-score for each flower class across the two models.

Table 15: Class-wise precision, recall and F1-score.

Model	Class	Precision	Recall	F1-score
Baseline	Daisy	0.9559	0.9630	0.9594
Baseline	Dandelion	0.9891	0.9579	0.9733
Baseline	Roses	0.9083	0.9083	0.9083
Baseline	Sunflowers	0.9182	0.9619	0.9395
Baseline	Tulips	0.9032	0.8750	0.8889
Channel–Spatial P3	Daisy	0.9333	0.9333	0.9333
Channel–Spatial P3	Dandelion	0.9208	0.9789	0.9490
Channel–Spatial P3	Roses	0.9167	0.9167	0.9167
Channel–Spatial P3	Sunflowers	0.9474	0.8571	0.9000
Channel–Spatial P3	Tulips	0.8800	0.9167	0.8980

25.4 Interpretation

The baseline achieved better overall accuracy and higher F1-scores for daisies, dandelions, and sunflowers. Channel–Spatial attention at P3 improved recall for dandelions and slightly improved F1-scores for roses and tulips, but its performance declined for other classes. Overall, the comparison shows that attention can affect flower classes differently, so confusion matrices and class-wise metrics are useful for understanding performance beyond overall accuracy.

26. Qualitative Analysis of Attention

26.1 Attention and Feature Visualization

Attention maps show the responses produced by attention modules, while intermediate feature maps reveal activations from convolutional layers. Overlays help identify whether the model focuses on the flower or responds strongly to background regions.

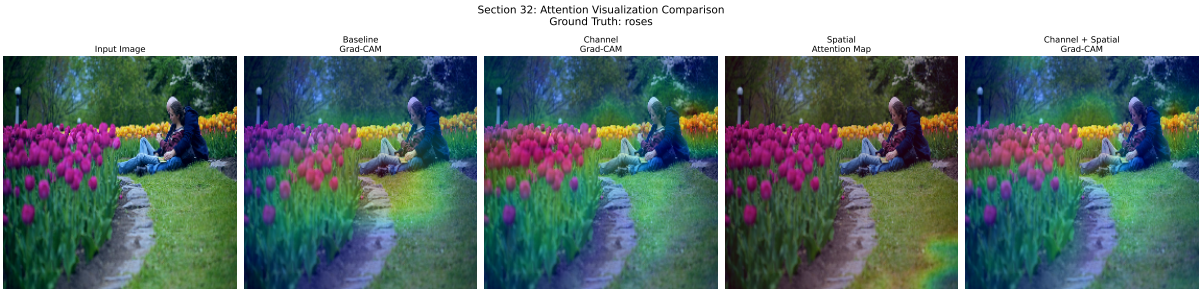


Figure 10: Combined visualization comparing selected attention models on the same flower image.

These visualizations allow the spatial responses of different models to be compared on the same image. However, highlighted regions should be interpreted carefully, as a heatmap alone does not prove that a prediction is correct or that the model has learned a reliable explanation.

26.2 Occlusion-Sensitivity Comparison

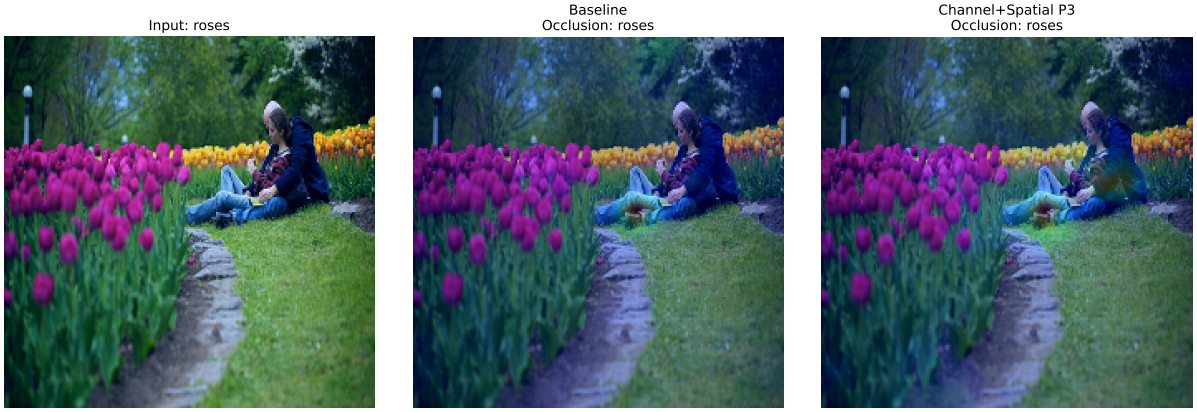


Figure 11: Occlusion-sensitivity comparison across the selected models.

Occlusion sensitivity measures how predictions change when parts of an image are hidden. A noticeable drop in the predicted class score suggests that the covered region contributes to the prediction. This complements attention maps, although it measures prediction sensitivity rather than attention weights directly.

26.3 Interpretation

The visual analysis focuses on whether models respond to the flower itself or to irrelevant background regions, and whether these patterns differ between correct and incorrect predictions. Together with accuracy and class-wise metrics, these visualizations provide a clearer picture of model behaviour on individual images. Any claim that an attention mechanism improves localization should be supported by the actual visual evidence.

27. Grad-CAM Visualization

27.1 Mathematical Formulation

Grad-CAM (Gradient-weighted Class Activation Mapping) highlights image regions that contribute to a model's prediction for a target class. Let A^k denote the k-th feature map of a convolutional layer and y^c the score for class c. The importance weight for feature map k is calculated as

$$\alpha_k^c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM heatmap is then computed as

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The resulting heatmap is resized to the input image dimensions and overlaid on the original image to show the regions most relevant to the selected class score.

27.2 Baseline and Attention Model Comparison



Figure 12: Grad-CAM visualization for the baseline ResNet50.

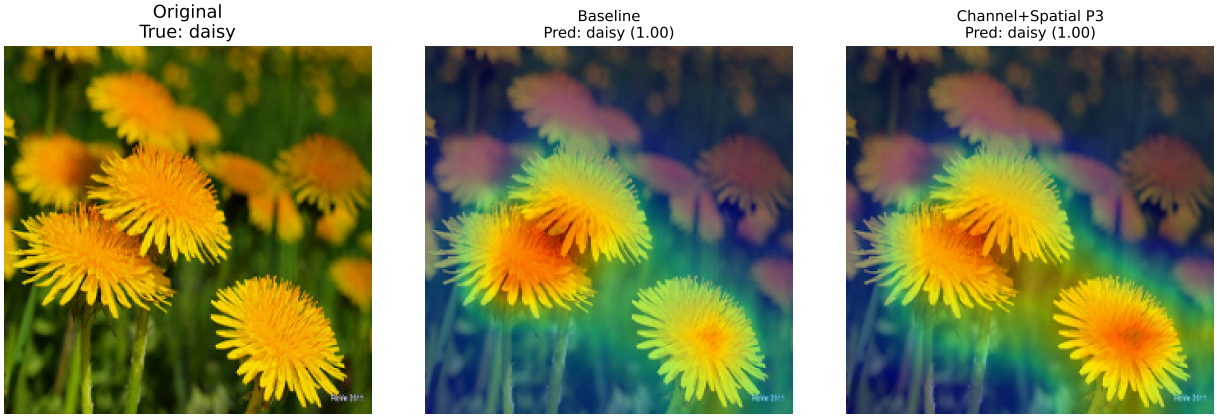


Figure 13: Grad-CAM comparison across the evaluated model variants.

For a fair comparison, the baseline and attention-based models should be evaluated using the same input image and target class. The heatmaps can then be examined to see whether the models focus on the flower, its petals, or surrounding leaves and background regions.

27.3 Interpretation

Grad-CAM helps explain which image regions influence a class prediction. Unlike internal attention maps, it uses gradients to highlight features relevant to the selected class. The visualizations should be interpreted alongside the model's predictions, since strong activation does not necessarily indicate correct localization. Comparing both correct and incorrect predictions can provide a clearer understanding of how the models use image features.

28. Internal Attention Maps

28.1 Channel-Attention Visualization

Channel attention assigns importance weights to feature channels, represented as

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Figure 14 shows the 20 highest channel-attention weights, allowing the relative importance of different feature channels to be compared.

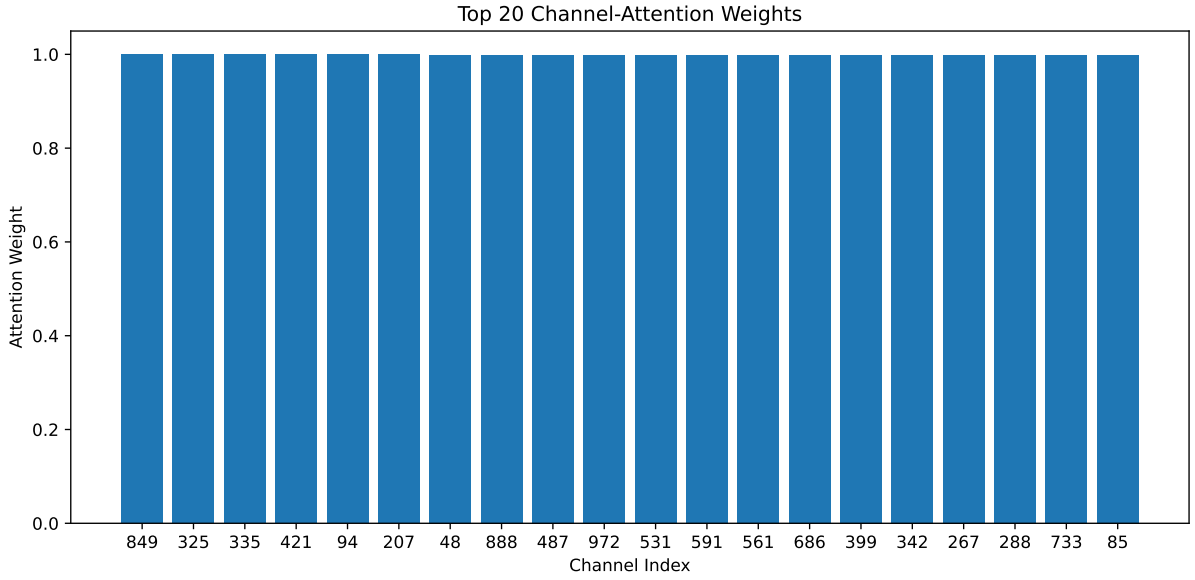


Figure 14: Top 20 channel-attention weights for the selected input image.

28.2 Spatial-Attention Visualization

Spatial attention assigns weights to locations in a feature map:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

The attention map is resized to the input resolution of 224×224 and overlaid on the original image to show which spatial regions receive stronger responses.

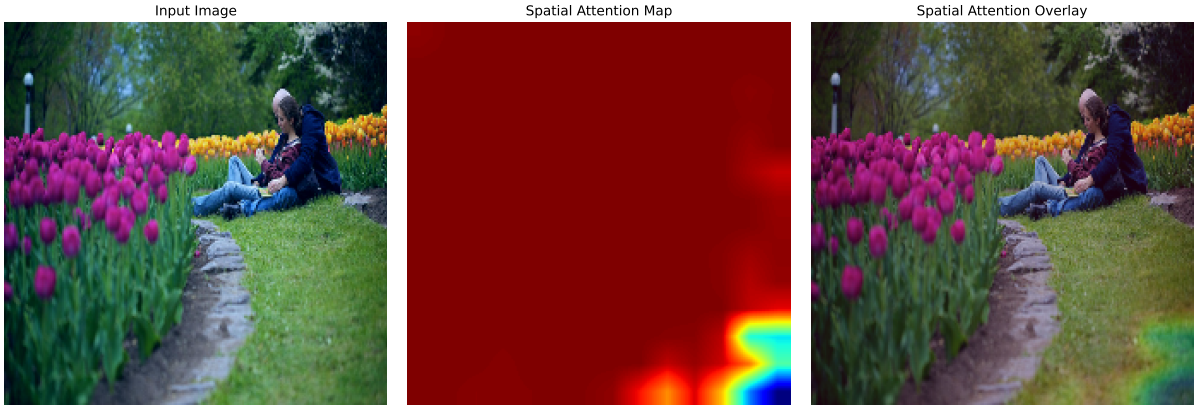


Figure 15: Input image, spatial-attention map, and attention overlay.

28.3 Self-Attention Visualization

Self-attention captures relationships between spatial positions in a feature representation. An aggregated attention response can be reshaped and visualized to examine how attention is distributed across the image.

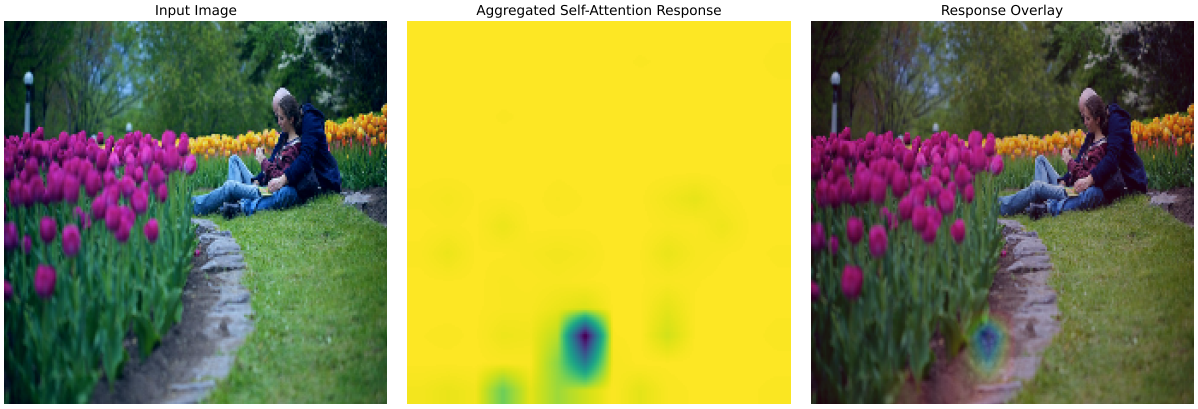


Figure 16: Aggregated self-attention response and its overlay.

28.4 Interpretation

These visualizations show attention from three perspectives: channel attention highlights the relative importance of feature channels, spatial attention shows how weights vary across locations, and self-attention represents relationships between spatial positions. Together, they help explore how the model processes an image, but the visual patterns alone do not prove that its predictions are correct or that the highlighted regions are causally responsible for them.

29. Attention Position Visualization

Spatial attention was compared at three ResNet50 stages: P2, P3, and P4. The same test image was used at each stage to examine how attention patterns change with feature depth.



Figure 17: Spatial-attention maps at stages P2, P3, and P4 for the same input image.

Interpretation

The visualizations allow attention patterns at different stages to be compared, from fine image details to broader spatial structures. The maps should be examined to identify which regions receive stronger attention at each stage. Any claim that a particular stage focuses more effectively on flower features should be supported by the visible results.

30. Qualitative Comparison Across Attention Mechanisms

30.1 Qualitative Model Comparison

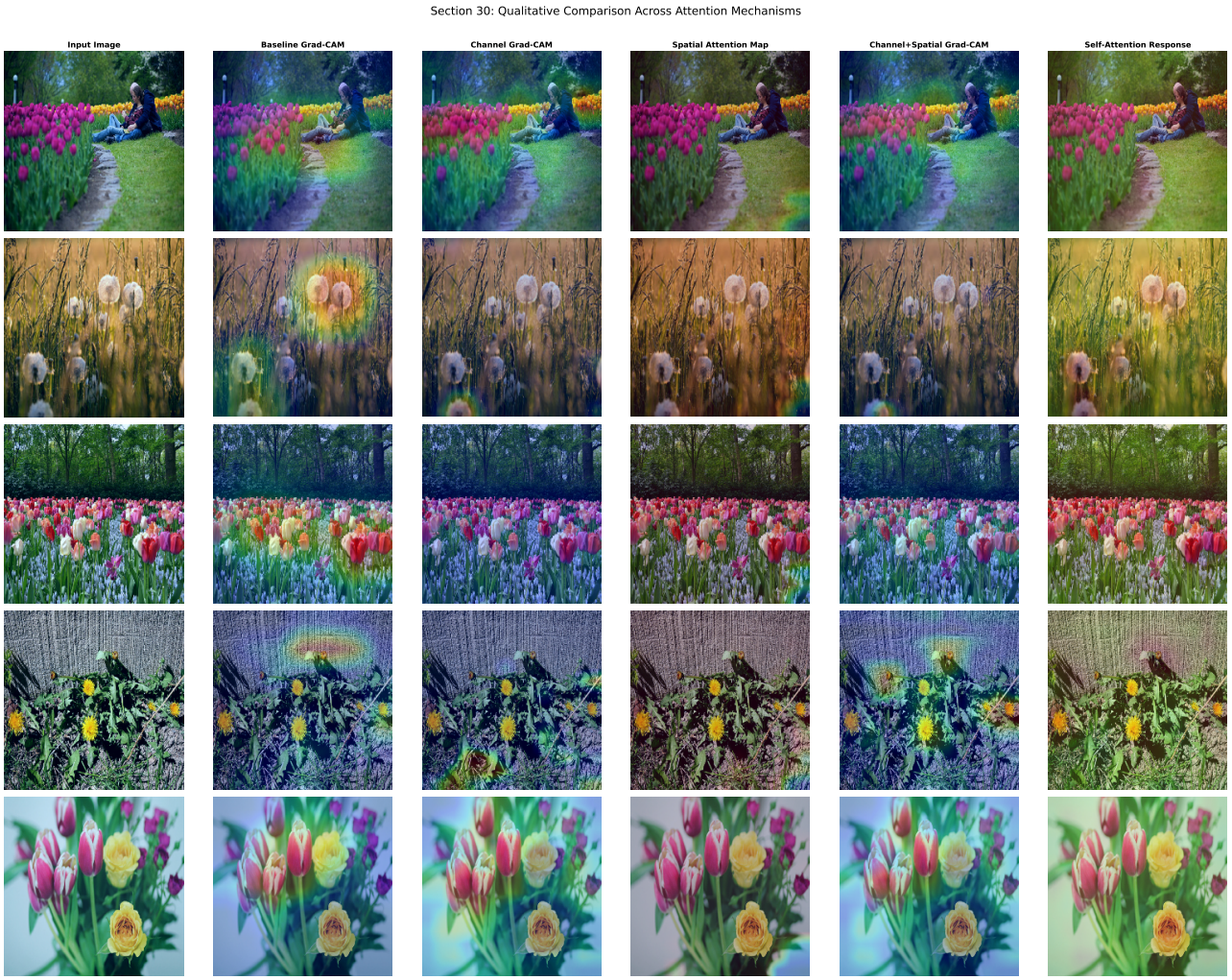


Figure 18: Qualitative comparison of baseline and attention-based models across selected flower images.

30.2 Prediction Analysis

Table 16 compares model predictions for two examples misclassified by the baseline.

Table 16: Predictions on examples misclassified by the baseline model

Model	True Class	Predicted Class	Confidence	Correct
Baseline	Daisy	Dandelion	0.4546	No
Channel Attention	Daisy	Daisy	0.9168	Yes
Spatial Attention	Daisy	Dandelion	0.8484	No
Channel+Spatial P3	Daisy	Dandelion	0.8968	No
Self-Attention	Daisy	Dandelion	0.4989	No
Baseline	Tulips	Roses	0.9319	No
Channel Attention	Tulips	Roses	0.9274	No
Spatial Attention	Tulips	Roses	0.9665	No
Channel+Spatial P3	Tulips	Roses	0.9961	No
Self-Attention	Tulips	Tulips	0.8585	Yes

For the daisy image, Channel Attention corrected the baseline's prediction, while the other models still classified it as dandelion. For the tulips image, Self-Attention made the correct prediction, whereas the remaining models predicted roses. Notably, Channel+Spatial P3 assigned 99.61% confidence to its incorrect prediction, showing that high confidence does not always mean a prediction is correct.

The three examples correctly classified by the baseline were also classified correctly by all four attention-based models, although their confidence levels differed.

30.3 Interpretation

The selected examples show that attention mechanisms can affect individual predictions differently. Channel Attention and Self-Attention each corrected a different baseline error, but neither consistently resolved both mistakes. These observations illustrate the potential benefits and limitations of attention, but broader evaluation is needed before drawing conclusions about overall model performance.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

32. Recommended Visualization Layout

32.1 Grad-CAM and Attention Visualization



Figure 19: Baseline Grad-CAM visualization for a representative test image.

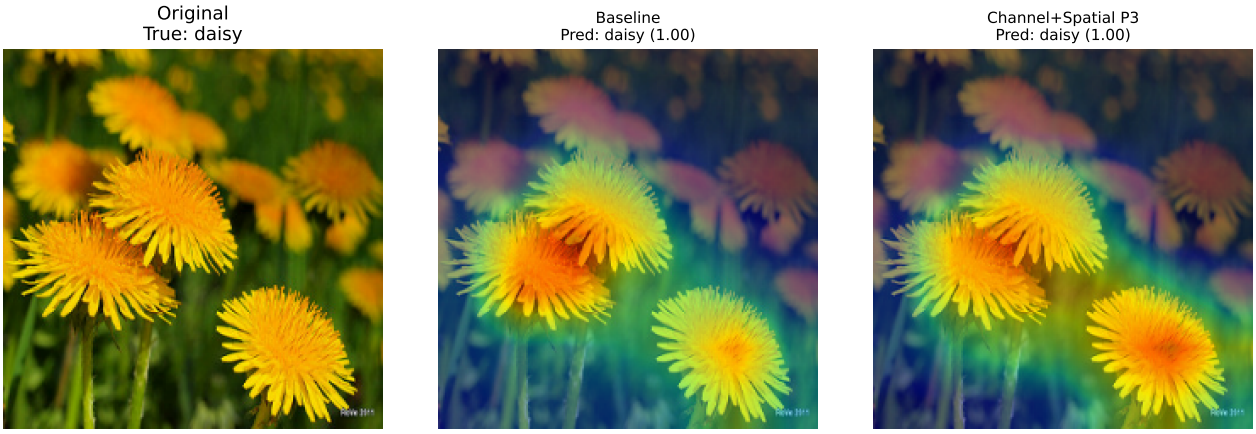


Figure 20: Grad-CAM comparison across the baseline and attention-based models.

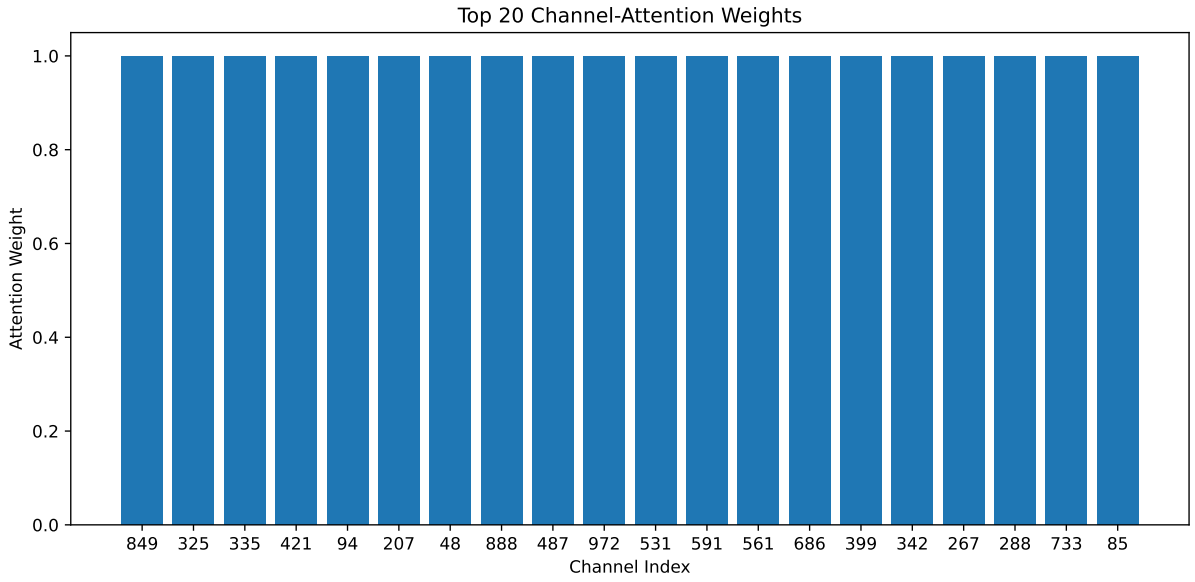


Figure 21: Channel-attention weights for the representative test image.

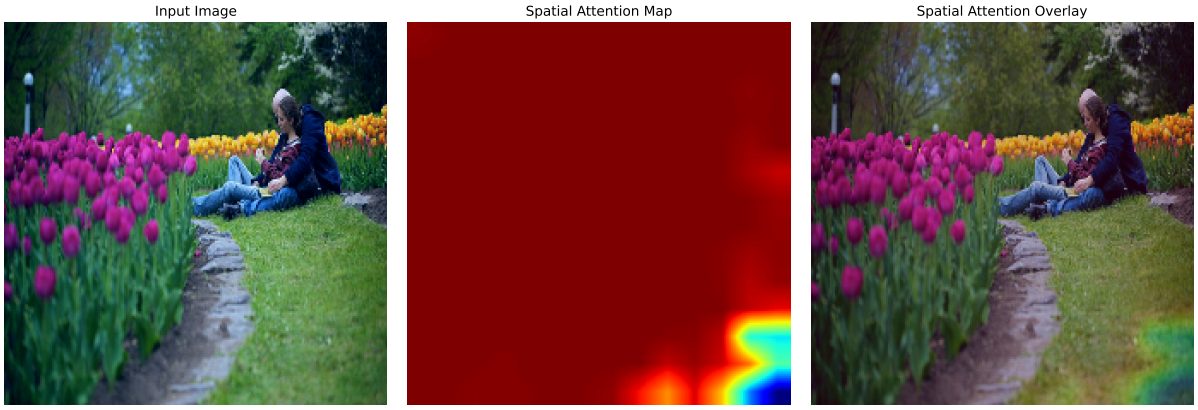


Figure 22: Spatial-attention map and overlay for the representative test image.

32.2 Interpretation

These visualizations offer different perspectives on how the models process an image. Grad-CAM highlights regions that contribute to a selected class score, while channel and spatial attention maps show the features and locations emphasized by the attention mechanisms. Comparing these outputs can help explain differences in model behaviour, but the visualizations should be interpreted alongside classification metrics rather than treated as proof of prediction accuracy.

33. Required Experimental Analysis

33.1 Quantitative Analysis

The models were compared using accuracy, precision, recall, and macro F1-score to evaluate overall classification performance.

Table 17: Quantitative comparison of the evaluated models.

Model	Accuracy	Precision	Recall	Macro F1
Baseline ResNet50	0.9347	0.9350	0.9332	0.9339
Channel Attention (P3)	0.9401	0.9409	0.9376	0.9389
Spatial Attention (P3)	0.9292	0.9275	0.9293	0.9283
Channel+Spatial (P2)	0.9111	0.9090	0.9151	0.9106
Channel+Spatial (P3)	0.9201	0.9196	0.9206	0.9194
Channel+Spatial (P4)	0.9220	0.9196	0.9222	0.9201
Self-Attention (P3)	0.9347	0.9360	0.9331	0.9343
SE (P3)	0.9310	0.9300	0.9312	0.9302
ECA (P3)	0.9201	0.9191	0.9226	0.9201

Channel Attention at P3 achieved the highest accuracy (94.01%) and macro F1-score (0.9389), outperforming the baseline accuracy of 93.47%. The results show that the effectiveness of attention depends on both the mechanism and its insertion position.

33.2 Complexity Analysis

Model complexity was assessed using total parameter count and average inference time.

Table 18: Parameter count and inference time of the evaluated models.

Model	Total Parameters	Inference Time (ms/image)
Baseline ResNet50	23,597,957	7.5673
Channel Attention (P3)	23,730,117	7.6675
Spatial Attention (P3)	23,598,056	7.6061
Channel+Spatial (P2)	23,631,368	7.8450
Channel+Spatial (P3)	23,730,216	7.6711
Channel+Spatial (P4)	24,124,520	7.7778
Self-Attention (P3)	24,909,958	7.8668
SE (P3)	23,730,117	7.5926
ECA (P3)	23,597,960	7.8403

The baseline had the lowest parameter count and fastest measured inference time, while Self-Attention had the highest parameter count and slowest inference time. Channel Attention achieved the strongest classification performance with only a modest increase in model size and inference time compared with the baseline.

33.3 Qualitative Analysis

Grad-CAM and internal attention maps were used to examine how the models responded to flower images. These visualizations help compare highlighted regions and explore differences in model behaviour.

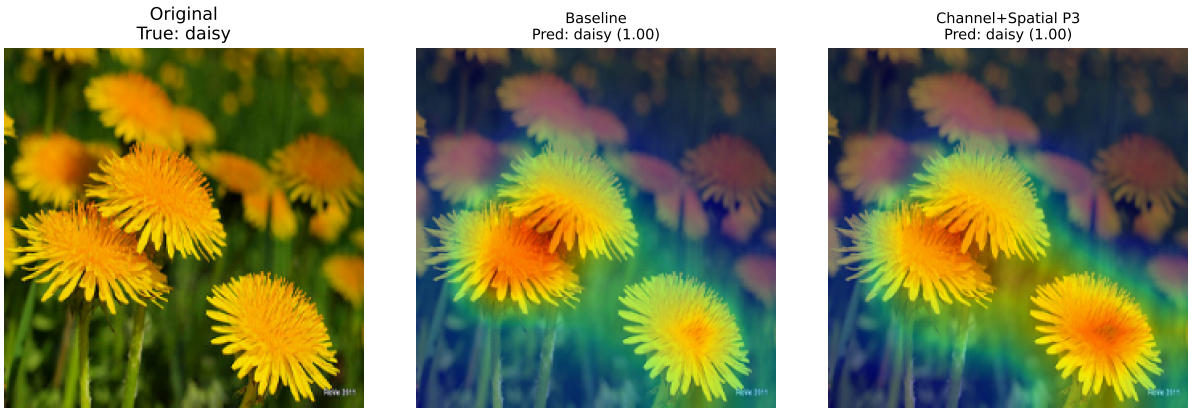


Figure 23: Grad-CAM comparison across the baseline and attention-based models.

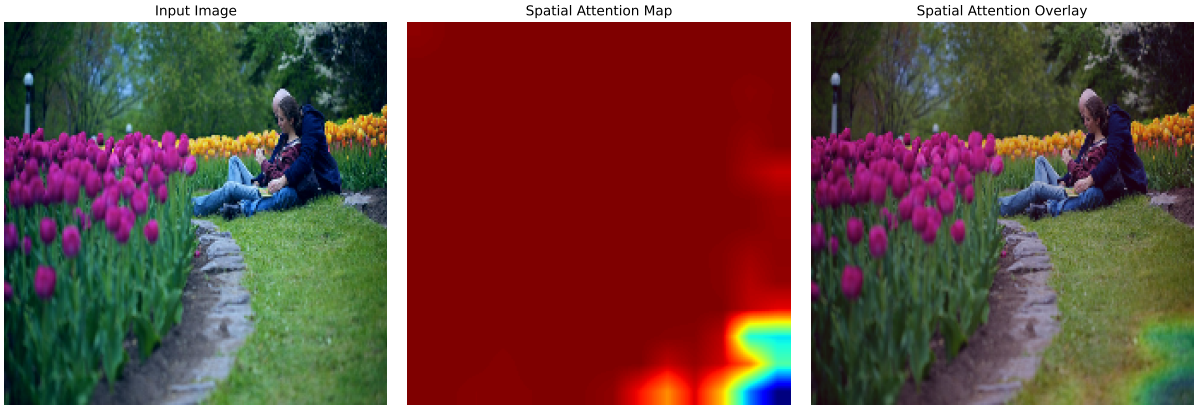


Figure 24: Spatial-attention map and overlay for a representative input image.

33.4 Interpretation

Overall, Channel Attention at P3 provided the best classification performance, while the baseline remained the fastest and smallest model in terms of total parameters. Self-Attention increased model complexity without improving accuracy over the baseline. The visualizations offer additional insight into model responses, but should be interpreted alongside quantitative results rather than as standalone evidence of better classification.

34. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?**
   Residual connections allow a block to learn a residual mapping rather than an entirely new transformation. They improve gradient flow and make deep neural networks easier to train.
2. **What does conv2_x represent compared with conv5_x?**
   The conv2_x stage extracts relatively low-level features, such as edges and textures, at higher spatial resolution. The conv5_x stage captures deeper, more semantic features at lower spatial resolution.
3. **Why does spatial resolution decrease with network depth?**
   Downsampling reduces the height and width of feature maps, helping control computational cost while increasing the effective receptive field.
4. **Why does the number of feature channels increase with network depth?**
   Deeper stages typically use more channels to represent increasingly complex patterns and semantic information.
5. **What are low-level, intermediate, and high-level features?**
   Low-level features represent edges and textures, intermediate features capture shapes and patterns, and high-level features encode semantic information useful for classification.
6. **What is the purpose of an attention mechanism?**
   Attention adaptively emphasizes informative features and reduces the contribution of less relevant information, potentially improving feature representations.
7. **What does channel attention attempt to learn?**
   Channel attention learns the relative importance of feature channels and assigns weights to emphasize informative ones.
8. **What does spatial attention attempt to learn?**
   Spatial attention learns which locations in a feature map are most relevant to the task.
9. **How is channel attention different from spatial attention?**
   Channel attention reweights feature channels, whereas spatial attention reweights locations across the height and width of a feature map.
10. **Why can channel and spatial attention be used sequentially?**
    Channel attention can first emphasize useful feature types, after which spatial attention highlights relevant locations within the refined feature maps.
11. **Why might channel attention behave differently at conv3_x and conv5_x?**
    These stages represent different levels of abstraction. conv3_x captures more local patterns, while conv5_x contains more semantic features, so the most useful channel weights may differ.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    Intermediate stages retain useful spatial detail while capturing meaningful patterns. Attention can highlight relevant regions before further downsampling removes spatial information.
13. **Why is self-attention computationally expensive at high spatial resolution?**
    Self-attention models relationships between spatial positions. For $N = H \times W$ positions, the attention matrix

contains N^2 entries, increasing memory and computational requirements.

14. **What happens to the size of the self-attention matrix when spatial resolution increases?**
    The number of entries grows quadratically with the number of spatial positions. Doubling both height and width produces four times as many positions and approximately sixteen times as many attention-matrix entries.
15. **How does attention performance vary when the insertion position changes?**
    Performance depends on the features available at each stage. In the recorded experiments, Channel+Spatial attention achieved accuracies of 91.11% at P2, 92.01% at P3, and 92.20% at P4.
16. **How does attention at P2 differ visually from attention at P4?**
    P2 operates on higher-resolution, lower-level features and can preserve fine spatial details. P4 operates on lower-resolution, more semantic features and may emphasize broader class-relevant patterns.
17. **Does applying attention at multiple stages always improve classification performance?**
    No. Additional attention can increase model complexity and computational cost without improving accuracy. Its benefits must be verified experimentally.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    Yes. The order determines which features are reweighted first and may produce different feature representations. The actual performance difference must be established through experiments.
19. **How does the spatial-attention kernel size influence the attention map?**
    A larger kernel captures information from a wider neighbourhood, while a smaller kernel focuses on more local patterns. The effect on accuracy and inference time depends on the task and must be evaluated experimentally.
20. **How does SE attention differ from ECA attention?**
    SE uses channel-wise pooling followed by a bottleneck transformation to generate channel weights. ECA uses a lightweight one-dimensional convolution to model local interactions between channels without dimensionality reduction.
21. **How does CBAM combine channel and spatial attention?**
    CBAM applies channel attention followed by spatial attention, refining feature maps along both channel and spatial dimensions.
22. **How does self-attention differ from channel attention?**
    Channel attention estimates the importance of feature channels. Self-attention models relationships between spatial positions or tokens, allowing information to be integrated across different locations.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
    No. More parameters do not guarantee better performance. In the recorded experiments, self-attention had the largest parameter count, while Channel Attention at P3 achieved the highest accuracy.
24. **What is the difference between an internal attention map and Grad-CAM?**
    An internal attention map shows weights generated by an attention module. Grad-CAM uses gradients of a target class score with respect to feature maps to produce a class-specific localization heatmap.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
    Grad-CAM generates explanations after training by analysing a model's activations and gradients. The model does not need to be trained specifically to produce Grad-CAM visualizations.
26. **Does the attention model suppress irrelevant background regions?**
    Attention may reduce the contribution of less relevant regions, but it does not guarantee background suppression. The visualizations must be examined to determine whether this occurs in the tested images.
27. **Are attention maps different for correctly and incorrectly classified images?**
    They can differ because a model may focus on different features or regions. Comparing correct and incorrect predictions can help reveal these differences, although the maps alone do not explain every classification error.
28. **Which flower classes benefit most from attention?**
    The effect varies by class. In the recorded results, Channel+Spatial attention at P3 increased Dandelion recall from 95.79% to 97.89%, but reduced Sunflower recall from 96.19% to 85.71% compared with the baseline. Attention therefore does not benefit every class equally.
29. **What information does a confusion matrix provide beyond overall accuracy?**
    A confusion matrix shows which classes are correctly classified and which class pairs are frequently confused. It reveals class-specific error patterns that overall accuracy cannot show.
30. **Why is macro F1 useful in addition to accuracy?**
    Macro F1 calculates the F1-score for each class and averages the scores equally. It helps reveal differences in class-wise performance that overall accuracy may conceal.
31. **What is the accuracy–complexity trade-off?**
    It describes the balance between classification accuracy and computational requirements, including parameter count, memory usage, and inference time.
32. **What is the F1–inference-time trade-off?**
    It compares macro F1, which gives equal weight to each class, with the time required to make predictions. A small improvement in F1 may not justify a substantial increase in inference time.
33. **Why must every ablation study use the same dataset split?**

Using identical dataset splits ensures that models are evaluated on comparable samples. Different splits can introduce data-distribution differences that make the results less directly comparable.

34. **Why must the same training protocol be maintained across attention mechanisms?**
Consistent training settings reduce confounding factors, making it easier to attribute observed performance differences to the attention mechanism being investigated.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
Grad-CAM can reveal whether different models focus on similar or different image regions when predicting a class. These visualizations provide qualitative evidence about model behaviour, but they do not independently prove that an attention mechanism improves accuracy or provides a fully faithful explanation.

35. References

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